SESSION 1 — AFTERNOON

Statistics & Probability

The quantitative toolkit for all investment decisions

Session Roadmap — 2 Hours

Six sections. Each builds on the previous. Statistics is the language of investment risk and return.

01 **Data & Frequency**
Data types • frequency distributions • histograms • EAA Equity Index from CFA LM3

02 **Central Tendency**
Arithmetic • Fund Y geometric mean 14.9% (CFA LM1) • DCA harmonic EUR 76.48 • MWR vs TWR

03 **Dispersion**
MAD • std dev 2.53% (CFA LM3) • target semideviation 1.5% (CFA) • CV • Sharpe ratio

04 **Probability**
Basic rules • conditional • BankCorp/defaulted bond prob trees (CFA LM4) • Bayes

05 **Probability Distributions**
Normal • z-scores • skewness (CFA EAA Index) • kurtosis • lognormal

06 **Correlation & Regression**
Covariance • correlation • Anscombe quartet (CFA LM3) • spurious correlation • OLS

Learning Objectives

By the end of this session you will be able to:

- **Classify and organise data**
  Distinguish nominal, ordinal, interval and ratio scales. Build frequency distributions and interpret histograms.
- **Compute all return averages**
  Calculate arithmetic, geometric and harmonic means correctly. Apply Fund Y (14.9%) and DCA harmonic mean (EUR 76.48) from CFA LM1.
- **Measure dispersion**
  Compute variance, standard deviation (CFA: 2.53%), target semideviation (CFA: 1.5%) and coefficient of variation.
- **Apply probability rules**
  Use addition, multiplication and complement rules. Build probability trees. Apply CFA LM4 BankCorp and bond recovery examples.
- **Work with distributions**
  Use normal distribution and z-scores. Interpret skewness (CFA EAA Index: -0.426) and kurtosis (excess: 3.796).
- **Measure correlation**
  Calculate covariance and correlation. Identify Anscombe quartet pitfalls. Apply simple linear regression (OLS).

SECTION 01

Data & Frequency Distributions

Types of Data — Getting the Foundation Right

Before applying ANY statistical method, identify your data type. The type determines which statistics are valid and which are meaningless.

Nominal

Categories with no order or meaning. Only counts valid.

Examples:
Country of domicile.
Industry sector.
Bond currency.

Stats: Mode only.

Ordinal

Categories with a meaningful order, but unequal spacing.

Examples:
Credit rating (AAA > AA > A).
Analyst ranking 1–5.

Stats: Median, mode. NOT mean.

Interval

Equal spacing but NO true zero. Ratios meaningless.

Examples:
Calendar year (2020).
Temperature in Celsius.

Stats: Mean, std. Ratios invalid.

Ratio

Equal spacing AND true zero. Ratios fully meaningful.

Examples:
Return (%), price, EPS, market cap.

Stats: All statistics valid. Most financial data.

Practical implication

Frequency Distributions — CFA LM3: EAA Equity Index

A frequency distribution groups observations into intervals (bins). It is the first step in understanding a return series. From CFA Vol 1, LM3:

Daily Return Range (%)	Frequency	Cumulative %	Interpretation
Below −2.5%	12	~0.9%	Extreme losses — fat left tail
−2.5% to −0.8%	216	~17.5%	Moderate losses
−0.8% to 0%	493	~54.8%	Small losses, most common
0% to +0.9%	470	~90.9%	Small gains, second most common
+0.9% to +2.5%	64	~95.8%	Moderate gains
Above +2.5%	5	~100%	Extreme gains — thin right tail

CFA LM3 Key Observations

EAA Equity Index daily return sample: mean = 0.0347%, std = 0.8341%, skewness = −0.4260, excess kurtosis = 3.7962.
The histogram shows: (1) asymmetry — more extreme negative days than positive (negative skew), (2) more extreme days than normal distribution predicts (fat tails, positive excess kurtosis). Both are typical of equity return series.

Population vs Sample — Why the Distinction Matters

In finance we almost always work with SAMPLES (historical data) rather than the full population. This requires adjusted formulas.

Population (N)

All possible observations. Variance: $\sigma^2 = \Sigma(X_i-\mu)^2 / N$. Standard deviation: $\sigma = \sqrt{(\Sigma(X_i-\mu)^2/N)}$.
Mean: μ (Greek).
Use when you have the COMPLETE dataset.

Sample (n)

Subset of the population. Variance: $s^2 = \Sigma(X_i-\bar{X})^2 / (n-1)$.
Standard deviation: $s = \sqrt{(s^2)}$.
Mean: $\bar{X}$ (x-bar).
Divide by (n−1) not n: Bessel correction prevents underestimation.

WHY divide by (n−1)?

Dividing by n gives a biased estimate that systematically UNDERESTIMATES the population variance. Bessel's correction (n−1) adjusts for the fact that the sample mean is calculated from the same data, introducing a dependency that reduces one degree of freedom. For large n, the difference is negligible.

- **Descriptive statistics:** Describe the sample you have. Mean, std dev, percentiles. No inference to a larger population.
- **Inferential statistics:** Use the sample to make conclusions about the population. Hypothesis tests, confidence intervals.

SECTION 02

Central Tendency (CFA LM1 + LM3)

Arithmetic Mean — Properties and Limitations

$$\bar{X} = \Sigma X_i / n \quad \text{(sum of all observations divided by count)}$$

The arithmetic mean is the most common measure of central tendency, but it has important limitations that CFA emphasises.

Property	Description	Finance Application
Sensitivity to outliers	A single extreme value pulls the mean strongly.	P/E ratio analysis: one firm with P/E = 500 distorts mean. Use median instead.
Uses ALL data	Every observation contributes equally to the mean.	Returns: each year counts once, regardless of how much was invested.
Best for one-period	Correct expected return for a SINGLE future period.	Forward-looking return estimates, portfolio construction.
Do NOT use for multi-period	+100% then −50% = arithmetic mean 25% but you have LOST money.	Use geometric mean for multi-period compounding calculations.

CFA reminder: Trimmed and Winsorized Means

Trimmed mean: remove top/bottom k% of observations before averaging (e.g. remove top 5% and bottom 5%). Winsorized mean: replace extremes with the value of the nearest non-extreme observation. Both reduce outlier impact without losing observations completely.

Geometric Mean — CFA LM1 Fund Y: 14.9%

$$R_g = [(1+R_1)(1+R_2)\ldots(1+R_n)]^{(1/n)} - 1$$

The geometric mean gives the COMPOUND growth rate over multiple periods. It is ALWAYS less than or equal to the arithmetic mean.

FUND Y — directly from CFA Vol 1, LM1 Practice Problem 3
Five-year annual returns: Year 1 = +19.5%, Year 2 = −1.9%, Year 3 = +19.7%, Year 4 = +35.0%, Year 5 = +5.7%.

Step 1: Convert: 1.195 × 0.981 × 1.197 × 1.350 × 1.057 = 1.9841
Step 2: Fifth root: 1.9841^(1/5) = 1.1490
Step 3: Subtract 1: geometric mean = 14.9% per year (CFA answer)

Arithmetic mean = (19.5 − 1.9 + 19.7 + 35.0 + 5.7)/5 = 15.6% per year.
Difference = 0.7pp — not large here because volatility is moderate.

What 14.9% means: USD 1,000 × 1.149^5 = USD 1,984 — matches actual cumulative return.

When to use geometric mean

Measuring actual compound growth rate of investment over multiple periods.
Reporting historical performance

Relationship: Arith vs Geo

Arithmetic ≥ Geometric always.
Gap increases with return VOLATILITY.
High variance returns → larger gap

Harmonic Mean — CFA LM1 Dollar-Cost Averaging

$X^H = n / \Sigma(1/X_i)$ **Best for averaging prices or ratios applied to a fixed currency amount**

The harmonic mean is appropriate when investing a FIXED amount of money at different prices (cost averaging). From CFA LM1 Practice Problem 4:

CFA LM1 PRACTICE PROBLEM 4 — EUR 5,000 invested per year at prices 62, 76, 84, 90

A portfolio manager invests EUR 5,000 each year for 4 years at these prices per unit.
Question: What was the average price paid per unit?

Harmonic mean = 4 / [(1/62) + (1/76) + (1/84) + (1/90)]
= 4 / [0.01613 + 0.01316 + 0.01190 + 0.01111]
= 4 / 0.05230 = EUR 76.48 (CFA answer: A is correct)

Arithmetic mean = (62+76+84+90)/4 = EUR 78.00 — WRONG for this context.
Harmonic mean is lower because MORE units are bought at LOWER prices.
Year 1: 5000/62=80.6 units. Year 4: 5000/90=55.6 units. Early purchases dominate.

Key rule: Harmonic mean for price averaging when investing a fixed MONEY amount each period. Arithmetic mean only when buying a fixed number of UNITS each period.

Relationship: Arithmetic ≥ Geometric ≥ Harmonic (equal only when all values are identical)

Money-Weighted vs Time-Weighted Return (CFA LM1)

The same portfolio can produce different return measures. The CHOICE of measure depends on whether you are evaluating the INVESTOR or the MANAGER.

	Money-Weighted Return (MWR)	Time-Weighted Return (TWR)
Also called	Internal Rate of Return (IRR)	Geometric mean of sub-period returns
Captures timing?	YES — accounts for when cash invested	NO — removes timing effects
Large investment before crash	MWR << TWR (timing hurts)	TWR unchanged by cash flows
Large investment before rally	MWR >> TWR (timing helps)	TWR unchanged by cash flows
Best for evaluating	INVESTOR performance	MANAGER skill (CFA preferred)
Why TWR preferred by CFA	Manager cannot control client deposits/withdrawals	TWR isolates pure investment decisions
Practical limitation	Requires recording every cash flow	Requires portfolio values at each cash flow date

Median, Mode, Quartiles & Percentiles

When should you NOT use the arithmetic mean? When distributions are skewed or have significant outliers — use the median instead.

Median

Middle value in sorted data. NOT affected by outliers.

For n observations: position = (n+1)/2.
If n=11: median at position 6.
If n=10: median = avg of positions 5+6.

Use when: data is skewed (e.g. house prices, income, P/E ratios with outliers).

Mode

Most frequently observed value. Only measure valid for nominal data.

Distributions may be: unimodal (one peak), bimodal (two peaks), multimodal.

Finance: Mode is rarely used. The mode of a return distribution tells you the most common return, but not the expected return.

Quantiles (from CFA LM3)

MSCI World Index 10yr returns (sorted): −40.33, −5.02, 9.57, 10.02, 12.34, 15.25, 16.54, 20.65, 27.37, 30.79.
Question (CFA Practice Q6): 4th quintile = 80th percentile. L80 = (10+1)×(80/100) = 8.8.
Between 8th (20.65) and 9th (27.37): P80 = 20.65 + 0.8×(27.37−20.65) = 26.03%. CFA answer: B.
IQR = Q3 − Q1 = 3rd quartile value − 1st quartile value. Robust measure of spread.

SECTION 03

Dispersion (CFA LM3)

Range and Mean Absolute Deviation (MAD)

Before variance, two simpler measures of dispersion: Range (uses only 2 values) and MAD (uses all values, avoids the squared distortion).

$$\text{Range} = \text{Maximum} - \text{Minimum} \qquad \text{MAD} = \Sigma|X_i - \bar{X}| / n$$

EXAMPLE — Monthly Portfolio Returns (from CFA LM3 dataset): mean = 2.25%

Returns: 5, 3, −1, −4, 4, 2, 0, 4, 3, 0, 6, 5
Range = 6 − (−4) = 10 percentage points. Simple but uses only 2 observations.
MAD: |5−2.25| + |3−2.25| + |−1−2.25| + ... = 2.75+0.75+3.25+6.25+1.75+0.25+2.25+1.75+0.75+2.25+3.75+2.75 = 28.5/12 = 2.375%

MAD avoids squaring, making it easier to interpret (in the same units). But squaring is needed for variance formulas.

Range limitation

Sensitive to a single extreme outlier. If one year returns +200%, the range doubles.
Useful only for quick initial screening. Always report alongside std dev.

MAD vs Std Dev

MAD: average absolute deviation. More intuitive, less sensitive to extremes.
Std Dev: larger due to squaring. But std dev is needed for normal distribution properties.

Variance & Standard Deviation — CFA LM3 Practice Q7

$$s^2 = \Sigma(X_i - \bar{X})^2 / (n-1) \qquad s = \sqrt{(s^2)} \qquad \text{[divide by n−1 for sample]}$$

Directly from CFA Vol 1, LM3 Practice Problem 7. A 10-year return series. CFA asks for sample standard deviation.

CFA LM3 Practice Q7: 10-year fund returns

Returns: 4.5%, 6.0%, 1.5%, −2.0%, 0.0%, 4.5%, 3.5%, 2.5%, 5.5%, 4.0%
Mean = 30.0%/10 = 3.0% per year.

Selected deviations squared:
4.5: $(0.045-0.030)^2 = 0.000225$
6.0: $(0.060-0.030)^2 = 0.000900$
−2.0: $(-0.020-0.030)^2 = 0.002500$ ← largest contributor (most extreme negative)
5.5: $(0.055-0.030)^2 = 0.000625$

Sum of all 10 squared deviations = 0.005750
Sample variance $s^2 = 0.005750 / (10-1) = 0.000639$
Sample std dev $s = \sqrt{0.000639} = 2.53\%$ (CFA answer B: 2.5276% ≈ 2.53%).

Interpretation: returns typically deviated 2.53 percentage points from the 3.0% mean. Std dev is in the SAME units as the original data.

Target Downside Deviation — CFA LM3 Practice Q8

Investors care more about DOWNSIDE risk. Target semideviation measures dispersion BELOW a minimum acceptable return (MAR) only.

$$s_{target} = \sqrt{[\Sigma(X_i - B)^2 / (n-1)]} \quad \text{for all } X_i \leq B \text{ only}$$

CFA LM3 Practice Q8 — Same 10-year fund (mean 3.0%), target B = 2%

Only include years where return < 2%:

Year 3: return=1.5%. Deviation from 2%: −0.005. Squared: 0.000025

Year 4: return=−2.0%. Deviation: −0.040. Squared: 0.001600

Year 5: return=0.0%. Deviation: −0.020. Squared: 0.000400

All other years (4.5%, 6.0%, 4.5%, 3.5%, 2.5%, 5.5%, 4.0%) are above target → EXCLUDED.

Sum of squared deviations below target = 0.000025 + 0.001600 + 0.000400 = 0.002025

Target semideviation = √(0.002025 / 9) = √0.000225 = 1.50% (CFA answer B)

Interpretation: when the fund misses the 2% target, it misses by 1.5pp on average.

Std Dev = 2.53% vs Semidev = 1.50%

Semidev is lower because it only captures downside variability, not upside. Standard deviation penalises good performance equally with bad.

Sortino Ratio uses semideviation

Sortino = (Return − Target) / Target semideviation. It only penalises downside risk, unlike Sharpe which uses full std dev.

Coefficient of Variation — CFA LM3 Practice Q9

Standard deviation is not comparable across datasets with different means. CV solves this by expressing risk per unit of return.

$CV = s / \bar{X}$ (standard deviation divided by mean) Higher CV = more risk per unit of return

CFA LM3 Practice Q9 — Three Sectors

Utilities (UTIL): mean = 2.10%, std = 1.23%. CV = 1.23/2.10 = 0.59
Materials (MATR): mean = 1.25%, std = 1.35%. CV = 1.35/1.25 = 1.08 ← RISKIEST
Industrials (INDU): mean = 3.01%, std = 1.52%. CV = 1.52/3.01 = 0.51 ← LEAST risky per unit of return

CFA answer: B (Materials). Despite Industrials having the highest absolute std dev (1.52%), it also has the highest return (3.01%), making it the LEAST risky per unit of return.
Common mistake: confusing highest std dev with highest CV. Always compute the ratio.

Chebyshev Inequality: for ANY distribution, at least $(1 - 1/k^2)$ of observations lie within k standard deviations of the mean. k=2: at least 75%. k=3: at least 88.9%. Works even when distribution is not normal.

Sharpe Ratio — Risk-Adjusted Return per Unit of Total Risk

$$\text{Sharpe Ratio} = (R_p - Rf) / \sigma_p \quad \text{[excess return per unit of total volatility]}$$

William Sharpe (1966). The Sharpe ratio is the most widely used measure of portfolio performance. It rewards return above the risk-free rate, penalises total volatility.

EXAMPLE — Three Funds Compared

Risk-free rate = 3%. Which fund performed best on a risk-adjusted basis?

Fund A: Return=12%, σ=10%. Sharpe = (12−3)/10 = 0.90
Fund B: Return=16%, σ=18%. Sharpe = (16−3)/18 = 0.72
Fund C: Return=9%, σ=5%. Sharpe = (9−3)/5 = 1.20 ← BEST risk-adjusted

Fund B has highest raw return (16%) but highest risk (18%). Fund C earns the most return per unit of risk taken. Sharpe = 1.20 means 1.20% excess return per 1% volatility.

Limitation: Sharpe uses total volatility (σ), which includes upside. Sortino (which uses downside deviation only) is preferred when the return distribution is not symmetric.

In general: Sharpe ratio > 1.0 is good. > 2.0 is excellent. Negative = performed below risk-free rate.

SECTION 04

Probability (CFA LM4)

Basic Probability Rules

Probability provides the mathematical framework for quantifying uncertainty. Three axioms underpin all probability calculations.

Axiom 1	$0 \leq P(E) \leq 1$ for any event E	Probabilities are always between 0 (impossible) and 1 (certain). P(rain tomorrow) = 0.7 is valid. P(rain) = 1.5 is NOT valid.
Axiom 2	$P(\Omega) = 1$ (sample space probability = 1)	The sum of probabilities for ALL possible mutually exclusive outcomes = 1. Something MUST happen.
Complement	$P(E^c) = 1 - P(E)$	Probability of event NOT happening = 1 minus probability it does. P(no default) = 1 − P(default).
Addition	$P(A \cup B) = P(A)+P(B)-P(A \cap B)$	If events are MUTUALLY EXCLUSIVE (cannot both occur): $P(A \cup B)=P(A)+P(B)$. If not, subtract the overlap.
Multiplication	$P(A \cap B) = P(A \mid B) \times P(B)$	If events are INDEPENDENT: $P(A \cap B) = P(A) \times P(B)$. For dependent events, use conditional probability.

Conditional Probability — Updating on New Information

$P(A|B) = P(A \cap B) / P(B)$ [probability of A given B has already occurred]

Conditional probability is the foundation of Bayesian updating, which is central to investment analysis and risk management.

EXAMPLE — Credit Default

P(default) = 5%. P(negative earnings | default) = 80%. P(negative earnings | no default) = 20%.

P(negative earnings) = P(neg|default)×P(default) + P(neg|no default)×P(no default)

= 0.80×0.05 + 0.20×0.95 = 0.04 + 0.19 = 0.23

Bayes Formula: If company reports negative earnings, what is probability it will default?

P(default|neg earnings) = P(neg|default)×P(default) / P(neg earnings)

= (0.80 × 0.05) / 0.23 = 0.04 / 0.23 = 17.4%

Before seeing negative earnings: P(default) = 5%. AFTER seeing negative earnings: P(default) = 17.4%. Information UPDATES the probability.

Bayes Formula: $P(A|B) = [P(B|A) \times P(A)] / P(B)$ — Prior probability updated by new evidence = Posterior probability.

Expected Value — CFA LM4: BankCorp EPS

$$E(X) = \Sigma P(X_i) \times X_i \quad \text{[probability-weighted average of all outcomes]}$$

Directly from CFA Vol 1, LM4, Example 1. Expected value is a FORECAST, not a historical average. It can never be observed directly.

BANKCORP EPS — CFA LM4, Example 1

Probability distribution for BankCorp EPS this fiscal year:
P = 0.15, EPS = USD 2.60 | P = 0.45, EPS = USD 2.45 | P = 0.24, EPS = USD 2.20 | P = 0.16, EPS = USD 2.00
Check: 0.15 + 0.45 + 0.24 + 0.16 = 1.00 (probabilities sum to 1) ✓

E(EPS) = 0.15(2.60) + 0.45(2.45) + 0.24(2.20) + 0.16(2.00)
= 0.390 + 1.1025 + 0.528 + 0.320 = USD 2.34

Variance: $\sigma^2 = 0.15(2.60-2.34)^2 + 0.45(2.45-2.34)^2 + 0.24(2.20-2.34)^2 + 0.16(2.00-2.34)^2$
= 0.010140 + 0.005445 + 0.004704 + 0.018496 = 0.038785
Std dev = √0.038785 = USD 0.197 per share (CFA answer: ≈ 0.20)

Probability Trees — CFA LM4: Defaulted Bond Recovery

Probability trees organise conditional probabilities into a structured diagram. Directly from CFA Vol 1, LM4 Question Set:

DEFAULTED BOND RECOVERY TREE — CFA LM4

A defaulted bond has two economic scenarios for principal recovery:
Scenario 1: Probability = 0.75. Recovery either USD 0.90 (prob 0.45) or USD 0.80 (prob 0.55).
Scenario 2: Probability = 0.25. Recovery either USD 0.50 (prob 0.85) or USD 0.40 (prob 0.15).

Joint probabilities:
USD 0.90: 0.75 × 0.45 = 0.3375 | USD 0.80: 0.75 × 0.55 = 0.4125
USD 0.50: 0.25 × 0.85 = 0.2125 | USD 0.40: 0.25 × 0.15 = 0.0375 (Sum = 1.00)

E(recovery | Scenario 1) = 0.45(0.90) + 0.55(0.80) = USD 0.845
E(recovery | Scenario 2) = 0.85(0.50) + 0.15(0.40) = USD 0.485

E(recovery) = E(rec|S1)×P(S1) + E(rec|S2)×P(S2)
= 0.845 × 0.75 + 0.485 × 0.25 = 0.634 + 0.121 = USD 0.755

Total probability rule: $E(X) = \Sigma\, E(X|S_i) \times P(S_i)$ across all mutually exclusive and exhaustive scenarios.

SECTION 05

Probability Distributions

The Normal Distribution — Foundation of Quantitative Finance

$X \sim N(\mu, \sigma^2)$ Bell-shaped, symmetric, fully described by mean and variance

The normal distribution is the most important distribution in finance. Many risk models, option pricing formulas, and hypothesis tests assume normality.

Interval	Rule	Example: N(μ=10%, σ=15%)	Finance Application
μ ± 1σ	68.3% of observations	[−5%, +25%]	Captures most "normal" years
μ ± 2σ	95.5% of observations	[−20%, +40%]	95% confidence interval for annual return
μ ± 3σ	99.7% of observations	[−35%, +55%]	Extreme events still within model
Left tail	2.5% below μ−1.96σ	−19.4%	Value at Risk (5% VaR for one-year horizon
Standardised	Z = (X−μ)/σ	N(0,1)	Allows lookup in standard normal table

Key property: any linear combination of normal random variables is also normal. This makes portfolio analysis tractable.

Z-Scores — Standardising to N(0,1)

$$Z = (X - \mu) / \sigma \quad \text{[how many standard deviations from the mean]}$$

The z-score converts any normally distributed variable to the standard normal, allowing probability lookup in a z-table.

WORKED EXAMPLES

Fund A has normally distributed annual returns: $\mu = 8\%$, $\sigma = 12\%$.

Q1: What is the z-score for a return of −10%?
$Z = (-10\% - 8\%) / 12\% = -18/12 = -1.50$
$P(\text{return} < -10\%) = P(Z < -1.50) = 6.7\%$ (from z-table)

Q2: What is the probability of earning above 20%?
$Z = (20\% - 8\%) / 12\% = 12/12 = +1.00$
$P(\text{return} > 20\%) = P(Z > 1.00) = 1 - 0.8413 = 15.9\%$

Q3: What return corresponds to the 5th percentile (worst 5% of outcomes)?
From table: $P(Z < -1.645) = 5\%$. Return $= \mu + Z \times \sigma = 8\% + (-1.645)(12\%) = 8\% - 19.7\% = -11.7\%$
This is the one-year 5% Value at Risk for this fund.

Skewness — CFA LM3: EAA Equity Index

Skewness measures the asymmetry of a return distribution. It has direct implications for risk management and investor preferences.

Positive Skewness (right tail)

Mean > Median > Mode.
Long right tail: frequent small losses, occasional large gains.
Example: Venture capital, lottery tickets.

Investors LIKE positive skewness — large upside with limited downside.

Negative Skewness (left tail)

Mean < Median < Mode.
Long left tail: frequent small gains, occasional large losses.
Example: Short selling, selling options, many equity indices.

Investors DISLIKE negative skewness — risk of crash.

EAA EQUITY INDEX — CFA LM3, Example 6

Daily returns over 5 years: mean = 0.0347%, std dev = 0.8341%, skewness = −0.4260, excess kurtosis = 3.7962.

Negative skew (−0.4260): the index experiences more days with extreme LOSSES than extreme gains. The left tail is longer than the right. This is characteristic of most equity return series globally.
Implication: using only mean and variance to assess risk will UNDERESTIMATE the probability of large negative outcomes.

Kurtosis — Fat Tails in Financial Returns

Kurtosis measures the combined weight of the TAILS relative to the rest of the distribution. Financial returns consistently have fatter tails than the normal.

Excess Kurtosis = Kurtosis − 3 **Normal distribution has kurtosis = 3 and excess kurtosis = 0**

Type	Kurtosis	Excess Kurtosis	Tail Shape	Finance Implication
Leptokurtic (fat-tailed)	Above 3	Positive (> 0)	Fatter than normal	More extreme events than expected. Stock returns typical.
Mesokurtic	= 3	= 0	Normal	Normal distribution. Risk models work correctly.
Platykurtic (thin-tailed)	Below 3	Negative (< 0)	Thinner than normal	Fewer extreme events. Less common in finance.

CFA EAA Equity Index: Excess Kurtosis = +3.7962

The EAA Index has excess kurtosis of 3.7962 (CFA LM3). This means extreme daily returns (beyond ±2.5 standard deviations) occur MORE frequently than the normal distribution predicts.

Practical impact: if you use a normal-based VaR model and ignore fat tails, you will systematically UNDERESTIMATE tail risk. This was a major factor in the 2008 financial crisis when tail events were deemed 25-standard-deviation events — theoretically impossible under normality.

Lognormal Distribution — Why Prices Cannot Be Negative

Asset PRICES cannot go below zero. Asset RETURNS can be negative. This asymmetry means prices are modelled as lognormal, not normal.

Normal Distribution of Returns

Returns (%) can be negative.
Assumption: $R \sim N(\mu, \sigma^2)$
Used for: return forecasting, portfolio construction, Sharpe ratio.

Problem: if you compound large normal returns, prices can theoretically go negative.

Lognormal Distribution of Prices

If $\ln(P_t/P_{t-1})$ is normally distributed, then prices P_t are lognormally distributed.
Prices always ≥ 0 (bounded below at zero).

Used for: option pricing (Black-Scholes), price simulation, Monte Carlo.

RELATIONSHIP between normal and lognormal
If $X \sim N(\mu, \sigma^2)$, then e^X is lognormally distributed.

For stock prices: $P_t = P_0 \times e^{(r_1 + r_2 + \ldots + r_\theta)}$, where r_i are continuously compounded returns.
This means: the CONTINUOUSLY COMPOUNDED return $R = \ln(P_t/P_0)$ is normally distributed.
The DISCRETE return $(P_t/P_0 - 1)$ is lognormally distributed (hence always > -1, i.e. price never goes negative).

Key number: $E(P_t/P_0) = \exp(\mu + \sigma^2/2)$ for lognormal. The variance ADDS to expected price.

SECTION 06

Correlation & Regression

Covariance and Correlation — Measuring Co-movement

How do two variables move together? Covariance measures direction; correlation standardises it to [−1, +1] for comparability.

$$\text{Correlation } \rho = \text{Cov}(X,Y) / [\sigma(X) \times \sigma(Y)] \qquad \text{Range: } -1 \leq \rho \leq +1$$

Correlation ρ	Meaning	Finance Example
+1.0	Perfect positive. Move in lockstep.	Two stocks in same sector, same beta
+0.5 to +1.0	Strong positive. Tend to rise/fall together.	Diversified equity portfolio stocks
0	No linear relationship.	Uncorrelated assets: ideal for diversification
−0.5 to 0	Moderate negative. Tend to move opposite.	Stock and 10yr government bond (often)
−1.0	Perfect negative. One always rises when other falls.	Long/short hedge: mirror image

Diversification principle

Portfolio variance = $w^2_1\sigma^2_1 + w^2_2\sigma^2_2 + 2w_1w_2\sigma_1\sigma_2\rho_{12}$. When ρ<1, portfolio std dev < weighted average of individual std devs — the benefit of diversification. If ρ=−1 (perfect hedge): portfolio can achieve zero variance.

Anscombe's Quartet — CFA LM3: Always Plot the Data

Francis Anscombe (1973) created four datasets with identical statistics but completely different patterns. CFA LM3 presents this as a critical lesson about correlation.

CFA LM3: THE FOUR ANSCOMBE DATASETS — All have IDENTICAL summary statistics

All four datasets have: Mean(X) = 9.0, Mean(Y) = 7.5, Var(X) = 11.0, Std dev(X) = 3.32, Std dev(Y) = 2.03, Correlation = 0.816.

Yet visually:
Dataset I: Standard linear scatter — linear model is appropriate.
Dataset II: Clear curved (quadratic) pattern — linear model is WRONG.
Dataset III: Perfect line with one extreme outlier — outlier drives the correlation.
Dataset IV: Vertical cluster at X=8 except one point at X=19 — correlation entirely driven by ONE observation.

Conclusion: ALWAYS visualise the data before relying on summary statistics. Same correlation = 0.816 can hide completely different underlying relationships.

CFA rule: Correlation alone is INSUFFICIENT. Always examine scatter plots, check for outliers, and verify that the relationship is linear.

Outlier problem

One extreme point can dominate the correlation coefficient, making it appear strong when the relationship is actually weak for most data.

Spurious correlation

CO_2 concentrations and global beer sales both rise over time: ρ = 0.824. Both are driven by a third variable (economic growth). Correlation does NOT imply causation.

Simple Linear Regression — OLS

$$\hat{Y} = \alpha + \beta X \qquad \beta = Cov(X,Y) / Var(X) \qquad \alpha = \bar{Y} - \beta\bar{X}$$

OLS (Ordinary Least Squares) minimises the sum of squared residuals. In finance: Y = returns of stock/portfolio, X = factor (market return, earnings, etc.)

EXAMPLE — Estimating Market Beta

Monthly data: Market excess return X, Stock excess return Y. n=24 months.
Cov(X,Y) = 0.0180 (=1.80% in decimal terms). Var(X) = 0.0120.

β = Cov(X,Y) / Var(X) = 0.0180 / 0.0120 = 1.50
$\alpha = \bar{Y} - \beta \times \bar{X}$. If $\bar{Y}$ = 1.2%/month and $\bar{X}$ = 0.7%/month:
α = 1.2% − 1.5 × 0.7% = 1.2% − 1.05% = 0.15% (Jensen's alpha)

Interpretation: β = 1.50 means the stock moves 1.5% for every 1% market move. If market rises 2%, stock is expected to rise 3%. α = 0.15% means risk-adjusted outperformance of 0.15%/month.

Measure	Formula	What it tells you
R^2 (R-squared)	= ρ^2 (or SSR/SST)	% of Y variance explained by X. R^2=0.81 means X explains 81% of Y's movement.
Residual (e_i)	$Y_i - \hat{Y}_i$	Unexplained part. Check for randomness — patterns signal model misspecification.

Sampling, Biases & the Central Limit Theorem

All statistical inference relies on sampling. But samples can be biased. The Central Limit Theorem (CLT) is what makes large-sample inference possible.

Key Sampling Biases

Survivorship bias: only including funds that survived. Overstates historical returns.

Look-ahead bias: using data not available at the time of the decision.

Data-mining bias: finding patterns by repeatedly testing until one works.

Time-period bias: results depend on the specific period chosen.

Central Limit Theorem

For samples of $n \geq 30$ (or any n if population is normal): the sampling distribution of $\bar{X}$ is approximately normal with:

Mean = μ (true population mean)
Std Error = $\sigma / \sqrt{n}$

This is WHY we can make probability statements about sample means, regardless of the original distribution shape.

Survivorship Bias Example

Hedge fund database shows average annual return of 12% over 10 years. However, 40% of funds that started 10 years ago no longer exist (closed due to poor performance). If those failed funds had average return of −5%, the TRUE average return across ALL funds = 0.60×12% + 0.40×(−5%) = 5.2%. Survivorship bias overstated returns by nearly 7 percentage points.

PRACTICE PROBLEMS

Work through each before checking the answers on the next slide.

Q1	Returns: 15%, 22%, −8%, 31%, 5%. Calculate: (a) arithmetic mean, (b) geometric mean. Why do they differ?
Q2	CFA LM1: EUR 5,000/year invested for 4 years at prices 62, 76, 84, 90. What average price was paid? (CFA answer: EUR 76.48)
Q3	Fund returns: 4.5%, 6%, 1.5%, −2%, 0%, 4.5%, 3.5%, 2.5%, 5.5%, 4%. Mean=3.0%. Calculate standard deviation (CFA answer: 2.53%).
Q4	Fund Y (CFA LM1): returns 19.5%, −1.9%, 19.7%, 35.0%, 5.7%. Calculate geometric mean. (CFA answer: 14.9%)
Q5	Defaulted bond (CFA LM4): Scenario 1 (prob 0.75): 45% chance recover 0.90, 55% chance recover 0.80. Scenario 2 (prob 0.25): 85% chance 0.50, 15% chance 0.40. What is expected recovery?

Practice — Worked Solutions

A1

Arith mean = (15+22−8+31+5)/5 = 65/5 = 13.0%. Geo mean = [(1.15)(1.22)(0.92)(1.31)(1.05)]^(1/5) − 1. Product = 1.15×1.22×0.92×1.31×1.05 = 1.8736. Geo mean = 1.8736^0.2 − 1 = 13.4% − wait: 1.8736^0.2 = 1.1337. Geo mean = 13.37% − 1 = 11.37%. Geo < Arith (13.0%) because of return volatility drag.

A2

Harmonic mean = 4 / [(1/62)+(1/76)+(1/84)+(1/90)] = 4 / 0.05230 = EUR 76.48. Arithmetic mean (wrong here) = (62+76+84+90)/4 = EUR 78.00. Use harmonic mean when investing a fixed CURRENCY amount at varying prices. Harmonic mean is lower because more units are purchased at lower prices.

A3

Mean=3.0%. Sum of squared deviations: (1.5)^2+3^2+(-0.5)^2+(-5)^2+(-3)^2+(1.5)^2+(0.5)^2+(-0.5)^2+(2.5)^2+(1)^2 = 2.25+9+0.25+25+9+2.25+0.25+0.25+6.25+1 = 55.5. Wait: deviations from 3: 1.5,3,-1.5,-5,-3,1.5,0.5,-0.5,2.5,1. Squared: 2.25+9+2.25+25+9+2.25+0.25+0.25+6.25+1=57.5. s^2=57.5/9=6.39%. s=√6.39=2.528% ≈ 2.53%. CFA answer confirmed.

A4

Product = 1.195×0.981×1.197×1.350×1.057. Calculate step by step: 1.195×0.981=1.172, ×1.197=1.403, ×1.350=1.894, ×1.057=2.002 (approximately). 5th root: 2.002^0.2 ≈ 1.149. Geo mean = 14.9%. CFA answer confirmed.

A5

E(rec|S1) = 0.45(0.90)+0.55(0.80) = 0.405+0.440 = USD 0.845. E(rec|S2) = 0.85(0.50)+0.15(0.40) = 0.425+0.060 = USD 0.485. E(rec) = 0.845×0.75 + 0.485×0.25 = 0.634+0.121 = USD 0.755 per USD 1 principal. CFA answer confirmed.

Formula Reference Card — Session 2

All formulas from this session in one place.

Arithmetic Mean	$\bar{X} = \Sigma X_i / n$ [forward-looking expected return]
Geometric Mean	$R^g = [(1+R_1)(1+R_2)\ldots(1+R_n)]^{(1/n)} - 1$ [compound growth]
Harmonic Mean	$X^H = n / \Sigma(1/X_i)$ [cost averaging, P/E]
Sample Variance	$s^2 = \Sigma(X_i - \bar{X})^2 / (n-1)$
Target Semideviation	$s_t = \sqrt{[\Sigma(X_i - B)^2 / (n-1)]}$ for all $X_i \leq B$ only
Coefficient of Variation	$CV = s / \bar{X}$ [risk per unit of return]
Sharpe Ratio	$Sharpe = (R_p - Rf) / \sigma_p$
Expected Value	$E(X) = \Sigma P(X_i) \times X_i$
Conditional Probability	$P(A\|B) = P(A \cap B) / P(B)$
Total Probability Rule	$E(X) = \Sigma E(X\|S_i) \times P(S_i)$
Bayes Formula	$P(A\|B) = [P(B\|A) \times P(A)] / P(B)$
Z-score	$Z = (X - \mu) / \sigma$
Correlation	$r = Cov(X,Y) / [\sigma(X) \times \sigma(Y)]$ Range [−1, +1]
OLS Slope	$b = Cov(X,Y) / Var(X)$
R-squared	$R^2 = \rho^2$ = % of variance in Y explained by X

KEY TAKEAWAYS

Session 2 — Statistics & Probability

- Fund Y (CFA LM1): geo mean = 14.9% for 5-yr returns 19.5/−1.9/19.7/35.0/5.7. Geo < arith (15.6%) due to volatility.
- DCA harmonic mean (CFA LM1): EUR 5k/yr at 62/76/84/90 → EUR 76.48 per unit (not arithmetic 78).
- CFA LM3 fund std dev: mean=3.0%, 10 years → s = 2.53%. Target semidev at 2% = 1.50%.
- CV (CFA LM3): Materials riskiest (CV=1.08) despite not having highest std dev. Always compute the ratio.
- BankCorp (CFA LM4): E(EPS) = 0.15(2.60)+0.45(2.45)+0.24(2.20)+0.16(2.00) = USD 2.34.
- Defaulted bond (CFA LM4): E(recovery) = 0.75(0.845)+0.25(0.485) = USD 0.755 per USD 1.
- EAA Equity Index (CFA LM3): skewness=−0.426 (negative
- Anscombe (CFA LM3): four datasets, identical μ/σ/ρ